

SHATAYU AVINASH MEHTA

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SUMMARY

Robotics Engineer (M.S.) with a foundation in mechanical systems, specializing in the complete robotics life-cycle from simulation and computer vision to hardware integration and autonomous control. Passionate about bridging the gap between software algorithms and physical hardware to build robust, field-ready autonomous systems and intelligent robotic components.

EDUCATION

University of Minnesota Twin Cities

Sept. 2024-Dec 2026

Master of Science in Robotics Technology.

GPA: 3.75/4

Relevant Coursework: Robotics, Principles Wearable Technology, Machine Learning, Robot Vision, Linear Control Systems, Deep Learning, Computer Vision.

K.J. Somaiya College of Engineering

July 2018-May 2022

BTech in Mechanical Engineering.

GPA: 3.4/4

TECHNICAL SKILLS

CAD & Analysis Software: SolidWorks, ANSYS (Fluent & Workbench), SolidWorks Simulation, CREO, AutoCAD, XFLR5.

Design & Manufacturing Principles: Finite Element Analysis (FEA), Thermal Design & Analysis, Computational Fluid Dynamics (CFD), Design for Manufacturability (DFM), DFMEA, GD&T, Bill of Materials (BOM), Rapid Prototyping (CNC, 3D Printing, Laser Cutting).

Programming & Software: Python, C++, C, PyTorch, TensorFlow, YOLOv10, MATLAB, ROS2, Docker, Linux, RoboDK, Mujoco, Gazebo, mavSDK, ISSAC SIM, DJI OSDK, Mission Planner, QGroundControl.

Fundamentals: SLAM, Sensor Fusion, Computer Vision, Machine Learning, Deep Learning, Feedback Control (PID), Robot Manipulation, Forward & Inverse Kinematics, Ackermann Steering, Visual Servoing, Stereo Vision, 3D Reconstruction, Mechatronics, E-textile Fabrication.

Hardware & Sensors: NVIDIA Jetson AGX, LiDAR, RTK GPS, Time-of-Flight (ToF) Sensors, Arduino, Raspberry Pi, ArduPilot, PX4, UR5e Cobot, Stepper & Servo Motors.

EXPERIENCE

Research & Development Engineer - Indrones Solution PVT. LTD.

Nov. 2022 – Nov. 2023

- Spearheaded FUJIN VTOL project from inception to testing of second prototype.
- Collaborated with the production team, applying Design for Manufacturability (DFM) principles to transition prototype systems into manufacturing, reducing assembly time by 1.5 weeks.
- Engineered and implemented a prototyping solution for EPP foam machining, successfully reducing manufacturing costs by 10%.
- Evaluated and tested powerhouse options using E-CALC for both VTOL and UAVs achieving a Time-of-flight increase by 25% for FUJIN VTOL, also established parameters for the aircraft prior to testing.
- Designed and prototyped components (SOLIDWORKS, FUSION 300), trained manufacturing staff on 3D printing, and refined aerodynamics using CFD (ANSYS Fluent) and flight data (XFLR5, Mission Planner) to improve coefficient of lift to 0.428.

Team Onyx India | Design Team Lead

May 2019-Aug. 2021

- Fronted the design team for TEAM ONYX INDIA in SAE Aero Design 2021. Fostered and advised the design team during SAE Aero design 2022.
- Engineered a self-gliding aircraft with a flight controller for autonomous landing, capable of auto-deploying from a mother ship.
- Executed market research and testing to identify flexible, durable, and cost-efficient material alternatives, reducing costs by 45%.

PROJECTS

Autonomous Vision-Based Weed Spraying Rover

(Jetson AGX, YOLOv10, ArduPilot, LiDAR, SLAM)

- Deployed YOLOv26 seg on an NVIDIA Jetson AGX for real-time row detection and using the segmented pointcloud data to cleanup LIDAR point cloud.
- Integrated RTK GPS for global waypoint navigation and fused LiDAR data with wheel encoders for SLAM, ensuring precise row-centering.
- Configured ArduPilot for autonomous locomotion, establishing Ackermann steering parameters and tuning complex PID control loops to stabilize navigation and dynamic positioning.
- Engineered a closed-loop control system utilizing Time-of-Flight (ToF) sensors and limit switches to dynamically adjust the sprayer boom height based on vision coordinates.
- Designed a localized delivery system mapping motor PWM signals to spray coverage diameters, enabling precise fluid deposition via servos and solenoid valves.

3D Object Reconstruction with VGGT & YOLO

(Python, PyTorch, VGGT, YOLO, Open3D)

- Engineered a 3D reconstruction pipeline leveraging the Visual Geometry Grounded Transformer (VGGT) to generate dense 3D point clouds from 2D input imagery.
- Integrated YOLO and SAM for real-time object detection and segmentation to isolate regions of interest, optimizing reconstruction focus.
- Validated the pipeline on custom datasets, successfully visualizing high-fidelity depth maps and point cloud representations.

Computer Vision-Based Tool Sorting

(UR5 Cobot, OpenCV, TensorFlow, Python, RoboDK)

- Built a robotic sorting system using a UR5 Cobot, identifying tools with a CNN (TensorFlow) and extracting 6DoF pose estimation via OpenCV (ArUco markers).
- Programmed the complete pick-and-place sequence, integrating the UR5, Robotiq Gripper, and force-torque sensors to achieve specific torque requirements during assembly.

Wearable Ring Mouse Development

(Arduino, Sensor Integration, Embedded C++, Python)

- Architected a wearable computing device using an Arduino Nano 33 BLE, integrating IMUs for swipe-to-tab gesture recognition and Hall effect sensors for scrolling.
- Created embedded C++ firmware to process sensor data and a Python receiver script to execute system-level inputs via BLE using the *pynput* library.